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FarmLink: A Digital Agricultural Marketplace for Traceable Produce Aggregation, Smart Allocation, and Farmer Settlement

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***Abstract*— *Many farmers struggle to sell their produce in larger markets and make satisfactory prices, due to multiple intermediaries in the supply chain and lower proportion of the final price that goes to the farmer. At the same time, commercial buyers have difficulties acquiring a steady supply of produce with satisfactory quality from individual farmers. This paper presents FarmLink, a digital agricultural marketplace that brings together farmers, Collection Centers (CCs) and commercial buyers on a single platform. Produce delivered to CC is received, weighed and graded, and a digital record of a batch of produce with a unique identifier is created, with the ability to trace it back to an individual farmer, thus enabling per-farmer traceability even after being mixed with other batches. The platform allows for partial batch sales to be made, with the unsold amount going back into inventory. The platform facilitates smart allocation of produce to buyer orders, with prioritization based on freshness, and farmer rating as a tie-breaker in the case of similar harvest dates. A digital farmer ledger is maintained of advances, pending amounts and final settlements due to farmers. The platform is implemented in React with Vite and Tailwind CSS for the frontend, Node.js and Express for the backend, PostgreSQL with Prisma ORM for the database, and uses JWT-based role authentication for security. By connecting the digital marketplace with the physical produce, FarmLink creates transparency and traceability of agricultural products from farm to table.*

***Keywords*—** *Agricultural Marketplace, FarmLink, Collection Center, Batch Traceability, Smart Allocation, Farmer Ledger, Inventory Management, Digital Agriculture.*

I. INTRODUCTION

Agriculture is the lifeline of many people and vital component of the economy. Small and medium-scale farmers however struggle to directly engage with larger buyers as their produce must pass through multiple layers of intermediaries before it reaches the commercial market, which reduces farmer incomes and complicates the supply chain.

Commercial buyers requiring large volumes of produce must aggregate from many individual farmers as there are

inconsistencies in quantity, quality, harvest time and availability. While physical aggregation resolves the volume problem, recording and tracking ownership and remaining inventory can become a challenge.

Digital tools can help address this issue by creating a shared space for farmer and buyer engagement. However, an agricultural marketplace cannot exist purely on the digital space, as the physical handling of the produce is equally important. The product must be received, weighed, graded, stored, tracked and ultimately dispatched to the final buyer. The digital system must therefore also record what occurs to the produce at the Collection Center.

FarmLink has been designed accordingly and aims to connect farmers, Collection Centers and commercial buyers via a single medium. The Collection Center serves as the intermediate link to the marketplace: as the produce is brought in by the farmer, the quantity and grade are recorded digitally and a digital batch is created that is still linked to the farmer, allowing for traceability even once aggregated. A further aspect of the system is inventory management: a batch of produce does not have to be sold as a whole. If only a portion is bought, the amount sold is removed from the inventory while the remainder remains available. Orders are matched to batches based on freshness, with farmer rating used if the harvest dates are similar, with a digital ledger maintaining farmer payment information including advances, pending and settled payments. The objective of FarmLink is therefore not simply to create an online marketplace, but provide a connected system for produce aggregation, traceability, inventory management, order fulfilment and farmer settlement.

A. Objectives

The objectives of the FarmLink project are to:

- Provide a digital platform connecting farmers and commercial buyers via Collection Centers.
- Record and maintain produce received at the Collection Centers.
- Create unique digital batches for traceability.
- Track the quantity supplied by an individual farmer even after physical aggregation.
- Allow partial sales and maintain the remaining quantity in a certain batch.

- Help buyers find produce according to crop, quantity, grade and delivery location.
- Allocate available produce considering freshness and farmer rating.
- Maintain a digital ledger for payments to farmers and transaction history.
- Improve visibility and coordination between farmers, CCs and buyers.

B. Existing System

In the conventional agricultural supply chain, farmers often rely on middlemen or local market channels for selling their produce with no visibility over who their produce is reaching or how their crop is priced and distributed. Small individual supply quantities also prevent them from bidding for bulk orders.

Digital agricultural marketplaces allow farmers to list produce and buyers to place orders, and to a certain extent, improve farming market access. However, just a marketplace is insufficient to solve the physical aggregation problem - after farmers supply their produce to a common location, the system should maintain information on the produce's sources (how much each farmer supplied in terms of quantity and grade) as well as its current status. Handling of partial orders is a limitation too - if a buyer places an order which is lower than the quantity supplied by a farmer, the remaining quantity should be tracked, and with no batch level inventory, it becomes challenging to accurately record this, especially if the buyer's requirement is fulfilled with combine produce of multiple farmers.

C. Proposed System

FarmLink proposes a three-sided agricultural marketplace linking Farmers, Collection Centers (CCs) and Commercial Buyers. Farmers supply produce to a Collection Center where it is received, weighed, inspected, graded and recorded, and a unique batch is created for the accepted produce containing details like the supplying farmer, crop, grade, quantity and other related transaction information thus enabling continuous connection between the physical produce and its sources. Commercial buyers may browse marketplace inventory and filter produce according to their requirements. Once an order is placed, the system identifies suitable batches for fulfillment, prioritizing freshness so that older suitable produce is allocated first; if there are multiple suitable batches with similar harvest dates, farmer rating is used as a tie-breaker. The system also allows partial batch sales, say a 60 kg batch being used to fulfill a 40 kg order, with just 40 kg marked as sold while the remaining 20 kg continues to be visible in the inventory thus enabling the batch to be traced until its lifecycle ends. A digital farmer ledger maintains advance payments, pending earnings, final settlements and transaction history, giving farmers insight into the financials of their supplied produce. By combining these functions, FarmLink seeks to make agricultural procurement organized while improving traceability, inventory visibility, order fulfillment and payment transparency.

II. LITERATURE REVIEW

Digital technologies have been increasingly used to enhance the marketing and distribution of agricultural produce. Prior work has explored electronic marketplaces, digital platforms, farmer participation, Farmer Producer Organisations (FPOs), and digital supply chains, generally finding that online platforms enhance market access, information sharing, and coordination while highlighting the complexities of adoption, trust, infrastructure, awareness, and the limited scale of individual farmers.

Studies looking at digital trade and electronic marketplaces provide the background to this work. Das et al. [1] explored eNAM and digital agricultural trade, finding that digital trading systems enhance the access to markets but do not address the physical aggregation and batch level tracking of produce on large scale. Argade et al.
[2] studied electronic marketplaces and stressed the importance of information sharing between participants, indicating that quantity, quality, availability, and delivery information must be maintained and shared to facilitate coordination. Raghavendra et al. [3] used survey based analysis to identify factors influencing farmer participation in electronic marketing, while Agnihotri and Bhattacharya
[4] found that trust and perceived usefulness strongly influence digital marketplace adoption implying that a farmer facing platform needs to be simple and transparent to gain reliance.

A group of related studies look at collective participation. Shroff [5] discussed how farmer collectives enhance the market access for marginal farmers, and Kumar and Kumar [6] looked at digitalization within the FPO business ecosystem. These are relevant to FarmLink as Collection Centers offer a structure through which the produce of multiple farmers may be aggregated while retaining the farmer level records.

Recent work has focused more directly on digital agricultural platforms. Devika et al. [7] proposed a mobile platform for direct market access, Mishra et al. [8] studied digital e-marketplaces and identified digital literacy as a key adoption factor, Nandhini et al. [9] studied digital marketing in agriculture and how it may enhance market reach, and Khan [10] explored agricultural e-commerce and the infrastructure challenges affecting implementation. A further set of studies look at auction and supply-chain models: Dave and Sajja [11] proposed an e-auction model for FPOs, Ravi Kumar et al. [12] showed that access to information influences participant participation in chilli e-markets, and Gummagolmath and Lakshmi [13] discussed FPO federation models addressing the constraints due to scale. Kumar et al. [14] identified market related constraints affecting FPO performance, Pendyala et al.
[15] stressed the importance of awareness for successful adoption of e-commerce among FPOs, Singh et al. [16] showed how better coordination improves the performance of e-agricultural supply chains, and Kumar and Parshant
[17] presented ARGOMARKET, a web based agricultural marketplace demonstrating the feasibility of direct digital trading between farmers and buyers.

Taken together, these works show that digital platforms can enhance the market access of agricultural produce and

improve communication between producers and buyers, but most concentrate on a single element online trading, farmer participation, FPO coordination, or market information, giving less attention to the physical handling of produce after its arrival at an aggregation point. Batch level traceability when produce from multiple farmers is combined, flexible handling of partial sales, freshness aware order fulfilment, and integrated farmer settlement remains largely unaddressed as an integrated system. This is where FarmLink focuses its attention: instead of treating a farmer's produce as a mere online listing, FarmLink creates a digital batch when the produce is received and accepted at the Collection Center, retains that batch linked to the farmer through storage, sale, allocation, and dispatch, supports partial batch sales while retaining the ability to track remaining quantity, apply freshness first allocation with farmer rating as a tie-breaker and maintain a farmer level ledger for payments tracking, combining market access with traceability, inventory control, order fulfilment and settlement management in one platform.

III. METHODOLOGY

FarmLink is a role-based digital marketplace designed to connect farmers, Collection Centers (CCs), and commercial buyers. The methodology focuses on maintaining a connection between the physical movement of agricultural produce and its corresponding digital records; instead of viewing produce only as an online listing, the system creates and manages digital batches from the point at which produce is received at the Collection Center.

A. System Workflow

The FarmLink workflow begins when a farmer supplies produce to a Collection Center. The received produce is weighed, inspected, and graded by the Collection Center, and once accepted, a unique digital batch is created containing details such as the farmer, crop, grade, quantity, and relevant timestamps. The accepted produce is then added to the Collection Center's inventory; since each batch remains associated with its supplying farmer, the system can determine the source and remaining quantity of the produce even after aggregation. When a buyer places an order, the system checks available inventory and selects suitable batches using a freshness-first approach, preferring produce with earlier harvest dates and using farmer rating as a tie-breaker when harvest dates are similar. After allocation, the Collection Center receives the required picking information and prepares the produce for dispatch; once the order is fulfilled, the corresponding quantities are updated and the farmer's payment records are adjusted, creating a continuous flow from produce receipt to final settlement.

The system follows a layered web architecture, illustrated in Fig. 3.1. The frontend is developed using React with Vite and Tailwind CSS, providing separate role-based interfaces for farmers, Collection Centers, and buyers. The backend is implemented using Node.js and Express, handling business logic, authentication, inventory operations, order processing, allocation, and payment related records, with Prisma ORM used to communicate with the PostgreSQL database. JWT-based authentication secures role-based access across all layers.

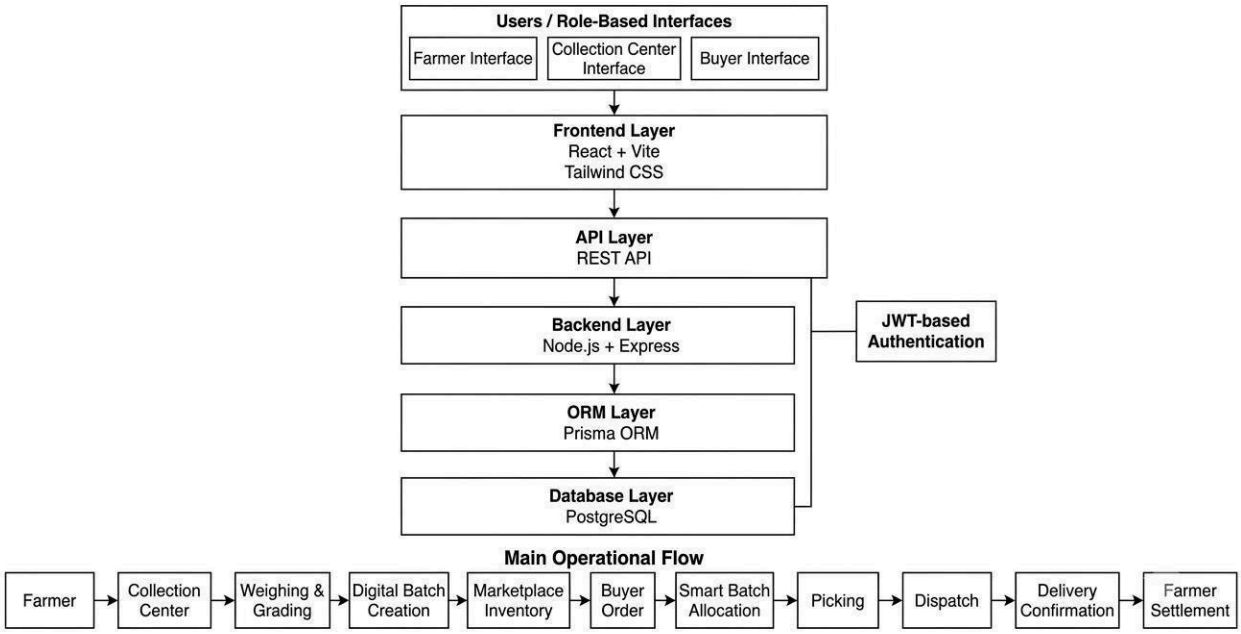


Figure 3.1: System Architecture of FarmLink

Fig. 3.1. System Architecture of FarmLink

As shown in the architecture, user requests flow from the role-based frontend, through a REST API to the Express backend, which applies business rules before reading from or writing to the PostgreSQL database through Prisma ORM. JWT-based authentication sits alongside the backend and API layers, verifying the role of each request so that farmers, Collection Center operators and buyers only access the functions relevant to their role. The lower

part of the figure traces the main operational flow of produce - from the farmer, through Collection Center inwarding, weighing and grading, digital batch creation, marketplace inventory, buyer ordering, smart batch allocation, picking, dispatch, delivery confirmation and farmer settlement - showing how each physical stage corresponds to an update in the digital record.

The operational sequence that underlies this architecture is represented in more detail as an activity flow, shown in Fig. 3.2. The activity begins with farmer produce submission and proceeds through Collection Center inwarding, weighing, grading, batch creation and inventory addition, prior to moving into buyer order placement, batch allocation, picking, dispatch and settlement.

Figure 3.2: Activity Diagram of FarmLink

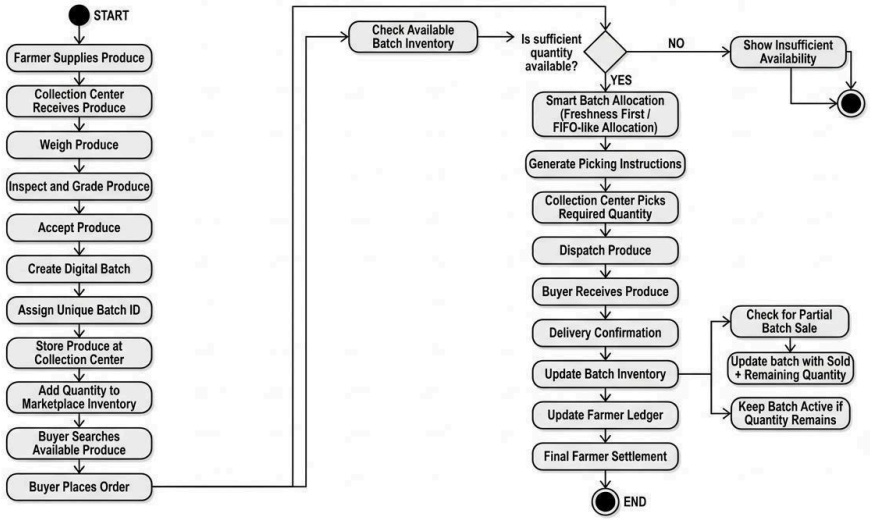


Fig. 3.2. Activity Diagram of FarmLink

As the activity diagram shows, once produce has been accepted and stored, it becomes available in marketplace inventory for buyers to search and order. When a buyer places an order, the system checks available batch inventory for sufficient quantity; if the available quantity is insufficient, the buyer is informed accordingly, and if sufficient produce is available, the system proceeds to smart batch allocation using the freshness-first rule described earlier. The Collection Center then receives picking instructions, dispatches the produce and the buyer's receipt and delivery confirmation trigger an update to batch inventory. If only part of a batch was required, the system checks for a partial sale, updates the batch with the sold and remaining quantities and keeps the batch active while quantity remains. This inventory update in turn updates the farmer ledger and completes the final farmer settlement for that transaction.

The high-level exchange of information between the main participants in this workflow is represented as a Level 0 data flow diagram, shown in Fig. 3.3.

Figure 3.3: Level 0 Data Flow Diagram of FarmLink

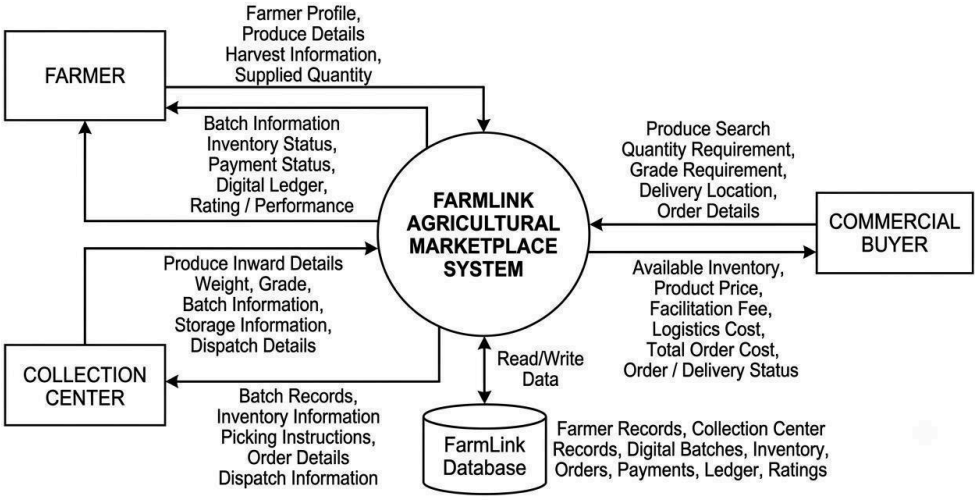


Fig. 3.3. Level 0 Data Flow Diagram of FarmLink

At this level, the Farmer, Collection Center and Commercial Buyer interact with the central FarmLink Agricultural Marketplace System, which in turn reads from and writes to the FarmLink database. Farmers supply profile, produce and harvest information and receive batch information, inventory status, payment status, ledger and rating updates in return. Collection Centers submit produce inward details, weight, grade, batch and storage information and receive batch records, inventory information, picking instructions and order and dispatch details. Commercial buyers submit produce search criteria, quantity, grade and delivery requirements and receive available inventory, pricing, facilitation and logistics costs, total order cost and order or delivery status. This high-level view sets the boundary of the system and the information exchanged across it, without detailing the internal processes that handle this exchange.

These internal processes are expanded in the Level 1 data flow diagram shown in Fig. 3.4.

Figure 3.4: Level 1 Data Flow Diagram of FarmLink

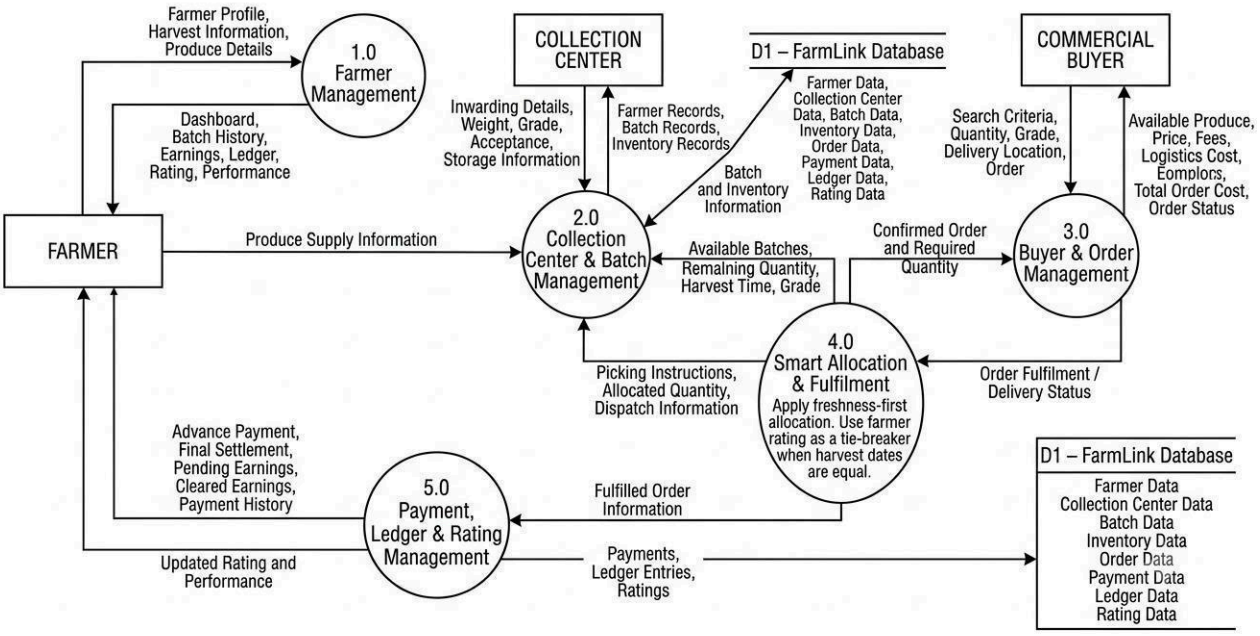


Figure 3.4: Level 1 Data Flow Diagram of FarmLink

Fig. 3.4. Level 1 Data Flow Diagram of FarmLink

The Level 1 data flow diagram divides the FarmLink system into five main processes: Farmer Management, Collection Center and Batch Management, Buyer and Order Management, Smart Allocation and Fulfilment and Payment, Ledger and Rating Management. The Farmer Management process records farmer profile and harvest information and returns dashboard, batch history, earnings, ledger and rating data. The Collection Center and Batch Management process receives inwarding, weight, grade and acceptance details from the Collection Center and produce supply information from farmers, and maintains batch and inventory records in the FarmLink database. The Buyer and Order Management process accepts search criteria and orders from buyers and returns available produce, pricing and order status, passing confirmed order and quantity requirements to the Smart Allocation and Fulfilment process. This process applies the freshness-first allocation rule described earlier, using farmer rating as a tie-breaker when harvest dates are equal, and returns picking instructions and dispatch information to the Collection Center while updating order fulfilment status for the buyer. Finally, the Payment, Ledger and Rating Management process updates the farmer ledger with advances, settlements and pending earnings once an order is fulfilled, and updates farmer rating and performance

based on buyer feedback. All five processes read from and write to the shared FarmLink database, which stores farmer, Collection Center, batch, inventory, order, payment, ledger and rating data.

B. Batch-Level Traceability

The first crucial component of the methodology is digitally tagged batches with associated supply farmer details and current status. An accepted quantity of produce is allocated a unique batch ID that contains information regarding the supplying farmer, crop, grade, quantity, harvest or inwarding details, and the current status. For example, a batch of tomato can be represented as:

Batch ID: BATCH-TOM-RAMAR-001
Farmer: Ramar | Crop: Tomato | Grade: A | Quantity: 60 kg

If a buyer buys 40 kg tomato from this batch, quantity is updated to 40 kg against ramar's account with 20 kg left in the system as the available stock, without affecting the batch traceability to the supplying farmer throughout the product life cycle.

C. Smart Allocation and Settlement

Farmlink makes fresh produce selection prioritized based on the harvest date, ensuring old produce does not get

stored where young produce gets dispatched first. In cases where the harvest date is the same or close, farmer rating gets prioritized and where quantities are limited, a smart selection can be made using multiple batch quantities to suffice an order against a single farmer category.

The methodology of settlements is broken down into two tranches with the first tranche representing an upfront payment made upon receiving, weighing, grading, and accepting the produce. The second tranche represents the counterpart of the settlement that is released either after selling the produce or completing the order. The proportion between the two can be configured based on business rules configured within the platform.

The farmer's ledger gets updated to record all transactions for reference, including cleared earnings, pending settlements, advances, final settlements, and history of transactions, providing a clear link between the produce supplied and the realized settlements in monetary terms.

The farmer's rating gets updated based on a buyer's feedback on the quality, grade specification accuracy, freshness, and general condition of the received produce. A weighted rating system is also applied to avoid spikes in ratings brought about by a single buyer's input, thus,

$$\text{New Rating} = (\text{Previous Rating} \times 0.7) + (\text{Latest Order Rating} \times 0.3)$$

this ensures ratings change gradually as new buyers continue to interact with the platform. It can be noted that the methodology combined batching while tracing to the farmer, inventory management, freshness prioritized allocation, order fulfillment, and digital settlement into a digital agricultural marketplace.

IV. PROJECT DESCRIPTION

A. System Description

FarmLink is implemented as a role-based agricultural marketplace which coordinates the activities of farmers, Collection Centers and commercial buyers, connecting the digital marketplace with the physical handling of agricultural produce so that produce can be tracked from the time it is received at a Collection Center until it is sold and dispatched. The workflow begins with the farmer supplying produce to a Collection Center, which records its quantity, checks its quality, assigns a grade and creates a digital batch linked to the supplying farmer. Batches are maintained as inventory, and when a buyer places an order, the system identifies suitable available batches based on crop, grade, quantity and freshness, giving preference to earlier harvested batches and using farmer rating when available batches have similar harvest dates. Partial sales are supported throughout, so that only the sold portion of a batch is deducted while the remainder continues to be available under the same batch record, preventing loss of traceability across multiple transactions. Payments are handled through the two-stage advance-and-settlement process described earlier, with all transactions reflected in the farmer's digital ledger.

B. Modules

FarmLink is organized into five major modules that manage the different activities involved in agricultural trading.

1) Farmer Management Module

This module maintains the profile and supply-related information of each farmer. Farmers can view their dashboard, supplied quantities, batch history, ratings, earnings and transaction details. The module also maintains the relationship between supplied batches, sales and farmer earnings, pending earnings, advances and final settlements, so that the produce they have supplied.

2) Collection Center and Batch Management Module

This module handles the physical inwarding and digital recording of produce. When produce reaches the Collection Center, it is weighed, inspected, graded and accepted before a batch is created; each batch receives a unique batch ID and contains details such as the farmer, crop, grade, quantity, and relevant timestamps. The module also maintains inventory and tracks the remaining quantity after sales, and the inventory can be viewed through a structured hierarchy such as Crop → Grade → Farmer → Batch → Remaining Quantity → Shelf-Life Status, making it easier for operators to locate and manage specific quantities.

3) Buyer and Order Management Module

This module gives commercial buyers access to available agricultural produce. Buyers can search and filter produce by crop, quantity, minimum grade and delivery location. Before confirming an order, the buyer can view the product price along with applicable platform facilitation and logistics costs; the system then records the order and passes the requirement to the allocation process.

4) Smart Allocation and Fulfilment Module

This module determines which available batches should be used to fulfil a buyer's order, applying freshness-first allocation and using farmer rating as a tie-breaker when harvest dates are similar. For example, if a buyer requires 100 kg of Grade A tomatoes and the available inventory consists of 60 kg from one farmer, 40 kg from another and 50 kg from a third, the system can allocate the first two batches to fulfill the order while leaving the third batch untouched. After allocation, the Collection Center receives the required picking information, and the selected produce is picked, prepared and dispatched to the buyer.

5) Payment, Ledger and Rating Module

This modu

This module maintains a ledger of financial information. Payments are recorded in two stages — an initial advance after accepted inwarding and a final settlement after the corresponding produce is sold or fulfilled — and a transaction history is maintained so that the relationship between supplied batches, sales and farmer earnings can be tracked. The rating component updates a farmer's rating based on buyer feedback on quality, grade accuracy, freshness and condition of the produce, using the weighted update described in Section III-C so that the system can give farmers a clear view of the money associated with their activity.

rating changes gradually rather than being dominated by a single recent order.

V. IMPLEMENTATION AND RESULT DISCUSSION

A. Implementation Results

FarmLink has been implemented as a web-based agricultural marketplace with distinct interfaces for farmers, Collection Centers, and commercial buyers. The implementation emphasizes maintaining consistent records across produce handling, inventory, orders, allocation, and farmer settlements.

1) Farmer Management

Farmer interface allows a view of their profile info, provided produce, batch history, ratings, and payment details, and their dashboard shows cleared and pending earnings. The digital ledger maintains the transaction history of the farmer's provided produce, allowing a view of their advances, final settlements, and pending amounts.

[Insert FarmLink Farmer Dashboard Screenshot Here]

2) Collection Center Inwarding

The Collection Center interface is used when produce is physically received from farmers. The operator records the crop, quantity, detail, and grade after inspection, and after acceptance, a corresponding digital record is made, which becomes the starting point of the produce tracking throughout the other marketplace workflow.

[Insert Collection Center Inwarding Interface Screenshot Here]

3) Digital Batch and Inventory Management

After the inwarding, the accepted produce is assigned a unique batch ID with information about the supplying farmer, crop, grade, quantity, and any relevant timestamps. The batch is added to the Collection Center inventory and since it is linked to the farmer, the system can identify the source of the produce even if produce from multiple farmers have been aggregated.

[Insert Digital Batch Creation and Inventory Management Screenshots Here]

4) Partial Batch Sales

FarmLink allows a partial sale of an available batch (e.g. 60 kg batch and order of 40 kg) by recording the 40 kg as sold and retaining the 20 kg as available inventory. The batch is active until its available quantity reaches 0, thus avoiding unnecessary duplicate records but still tracing the original produce.

[Insert Partial Batch Sale Screenshot Here]

5) Buyer Marketplace

The buyer interface shows available agricultural produce, allows searching according to their requirements, with filters including crop, quantity, minimum grade, and delivery location. The order summary breaks down the product price, platform facilitation fee, logistics cost, and total order cost before an order is confirmed.

[Insert Buyer Marketplace and Order Summary Screenshots Here]

6) Smart Batch Allocation

After placing an order, the system identifies suitable batches from the available inventory and prioritizes batches with earlier harvest dates. For cases with two or more batches with similar harvest dates, farmer rating is used as a tie-breaking factor, and the system can combine quantities from multiple batches when a batch is not enough to satisfy the complete order.

[Insert Smart Batch Allocation Screenshot Here]

7) Farmer Payment, Ledger, and Rating

The payment process records farmer earnings in two stages as described in Section III-C, and the farmer ledger provides a consolidated view of cleared, pending amounts, advances, and final settlements. Buyer feedback is used to maintain the farmer's performance rating, considering factors such as produce quality, grade accuracy, freshness, and condition upon arrival with the weighted update allowing the latest feedback to impact the rating without discarding their previous performance record.

[Insert Farmer Payment, Ledger, and Rating Screenshots Here]

8) Collection Center Outwarding

Once the required batches are allocated, the Collection Center prepares the produce for outwarding and the system provides the information needed to identify the selected batches and quantities. After the produce is dispatched, the corresponding inventory quantities are updated, thus completing the movement of produce from Collection Center inventory to the buyer.

[Insert Collection Center Outwarding and Dispatch Screenshot Here]

B. Discussion

The implementation showcases how FarmLink merges marketplace operations with physical produce management. A key design choice is the maintenance of digital records at the batch level rather than just on overall inventory to maintain farmer-level traceability even if produce from multiple farmers are aggregated at a Collection Center - directly addressing the gap identified in Section II, where existing digital marketplaces were found to be giving comparatively little attention to physical aggregation and batch-level tracking.

The partial-sale mechanism allows for more flexible inventory, since a batch does not need to be sold in one transaction and its remaining quantity can still participate in future orders. The freshness-first allocation approach is a practical way of managing the agricultural inventory, and the farmer rating mechanism provides an additional decision-making factor when batches have similar freshness. The two-stage payment process and digital ledger further enhance transparency by linking produce supply directly to farmer earnings. Since these features are part of the system's design rather than the subject of a separate field trial, they are presented here as implemented but not yet experimentally validated at scale; no performance figures, adoption statistics, or user-satisfaction measures are reported since none have been

collected. Overall, the implemented modules work together to provide a structured flow from farmer supply, through Collection Center handling, digital batch creation, marketplace ordering, batch allocation, and fulfilment, to farmer settlement.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

FarmLink introduces a digital agricultural marketplace to facilitate transactions between farmers, Collection Centers, and commercial buyers, emphasizing not only online buying and selling but also physical handling and tracking of agricultural produce. With digital batches, every quantity of produce maintains its association with its supplying farmer throughout the inventory and order process, and features such as partial batch sales, freshness-first allocation, farmer ratings, and Collection Center-based inventory management offer a more organized approach to agricultural fulfilment. The two-stage payment process and digital farmer ledger make it easier to track advances, pending earnings, and final settlements, and buyers are provided with structured information on produce availability, grade, quantity, pricing, and delivery-related costs. Overall, FarmLink aims to reduce the gap between farmers and commercial buyers by combining marketplace access, physical aggregation, batch-level traceability, inventory management, order fulfilment, and payment tracking within one system, providing a stronger foundation for transparent and coordinated agricultural transactions.

B. Future Work

The current system can be enhanced in several ways to improve its usability and scale. Future versions of FarmLink can include:

- QR and barcode-based batch tracking for faster identification of physical produce.
- Mobile applications for farmers, Collection Centers, and buyers.
- Real-time logistics tracking for better visibility during transportation.
- Demand and price prediction to support farmer and buyer planning decisions.
- Automated freshness monitoring using sensors or IoT devices for stored produce.
- Multi-language support to make the platform accessible to a wider range of users.
- Advanced analytics dashboards for monitoring farmer performance, inventory movement, sales, and demand.
- Scalable cloud deployment to support a larger number of farmers, Collection Centers, and buyers across regions.

These improvements can further enhance FarmLink as a scalable digital platform for agricultural supply-chain management.

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